

CRYPTOGRAPHY AND NETWORK SECURITY

PROJECT TOPICS, RULES & ACADEMIC GUIDELINES

Project Topics • Team Structure • Deliverables • Presentation & Viva

1. Project Guidelines

The project is designed to develop practical, analytical and research-oriented skills in Cryptography and Network Security. Each team will investigate one common problem and develop two technically meaningful approaches, followed by implementation, experimentation and comparative evaluation.

- Each project team must consist of 4–5 students.
- The team must be divided into two sub-groups, with 2–3 students in each sub-group.
- Both sub-groups must work on the SAME project/problem statement.
- Each sub-group must develop a DIFFERENT technical approach, technique, algorithm, architecture or methodology.
- The complete team must collectively analyse and compare both approaches and determine which approach is better and why.
- The comparison must be supported by suitable parameters such as security, performance, accuracy, computational complexity, processing time, resource utilization, scalability, reliability, usability or implementation complexity.

2. Topic Finalization

Students must select one topic from the approved list and report physically in class with the topic and team details:

Class	Topic & Team Finalization
CSE Tigers	1 September 2026
CSE Rhinos	4 September 2026

Come prepared with: selected topic, names of all team members, sub-group division, Approach 1 and Approach 2.

3. Deliverables

Sub-group	Mandatory Deliverables
2 Students	Working Project/Implementation; Academic Project Report; Project Presentation; Viva Voce
3 Students	Working Project/Implementation; Academic Project Report; Project Presentation; Viva Voce; DRAFT RESEARCH PAPER – MANDATORY

4. Mandatory Project Workflow

Problem Definition → Literature Review → Approach 1 → Implementation/Experimentation → Results → Approach 2 → Implementation/Experimentation → Results → Comparative Analysis → Best Approach → Technical Justification → Conclusion & Future Scope

5. Two-Approach Requirement

The two approaches must differ meaningfully in their underlying security methodology, algorithm, architecture, detection mechanism, cryptographic technique or analytical method. Merely changing the programming language, user interface, or making minor implementation changes will NOT be accepted as a second approach.

Example: For a Network Intrusion Detection project, Approach 1 could be signature-based detection and Approach 2 could be machine-learning-based anomaly detection. Compare measurable parameters and justify the better approach.

6. Academic Project Report Format

1. Title Page
2. Abstract (150–250 words)
3. Introduction
4. Problem Statement
5. Objectives
6. Literature Review
7. Existing System / Existing Approach
8. Proposed Approach 1
9. Proposed Approach 2
10. Methodology
11. System / Network Architecture
12. Implementation
13. Experimental Setup
14. Results
15. Comparative Analysis
16. Security Analysis
17. Discussion
18. Conclusion
19. Future Scope
20. References
21. Appendix, if applicable

7. Draft Research Paper – Mandatory for 3-Member Sub-Groups

The research paper must be based on the project and should include the research problem/motivation, literature review, research gap where applicable, methodology, technical contribution, experimental setup, results, comparison of both approaches, discussion, findings, conclusion, future scope and references.

8. Presentation & Viva

Project Presentations and Viva Voce will be conducted from 28 September 2026 to 1 October 2026.

- The complete project, implementation, academic report, presentation and research paper (where applicable) must be ready before the allotted presentation/Viva date.
- All team members must understand BOTH approaches. Viva questions may cover any part of the project.
- Students must be able to explain their individual contribution, technical decisions, algorithms, security implications, experimental results and comparative findings.

9. Important Academic Requirements

- Use credible academic and technical sources and cite them properly.
- Follow the prescribed academic report format and IEEE referencing style unless otherwise instructed.
- All figures, tables, graphs and screenshots must be numbered, captioned and discussed.
- Clearly distinguish information obtained from sources from the team's own analysis.
- The project must demonstrate implementation/experimentation wherever applicable, not merely a theoretical description.

- All team members must contribute meaningfully.
- Academic integrity must be maintained. Fabricated references, copied content and unsupported claims are unacceptable.
- Any delay or lack of preparedness from the students' end will result in deduction of marks.

10. Project Topics

Cryptographic Algorithms & Secure Data

No.	Project Topic	Suggested Two Approaches
1	AES vs ChaCha20 for Secure Data Encryption	AES-GCM vs ChaCha20-Poly1305
2	Secure File Encryption and Decryption System	AES-GCM vs ChaCha20-Poly1305
3	Image Encryption for Secure Storage and Transmission	AES vs ChaCha20
4	Secure Video Encryption Framework	AES-based vs selective encryption
5	Secure Database Encryption System	AES vs tokenization
6	Hybrid Encryption for Secure File Sharing	RSA-AES vs ECC-AES
7	AES Key-Size Performance Analysis	AES-128 vs AES-256
8	Secure Document Storage Using Hybrid Cryptography	RSA vs ECC key protection
9	Cryptographic Performance Analyzer	AES vs ChaCha20
10	Secure Backup Encryption Framework	Envelope encryption vs centralized key management
11	Block Cipher Mode Security Comparison	CBC vs GCM
12	Authenticated Encryption Framework	AES-GCM vs ChaCha20-Poly1305
13	Lightweight Cryptography for Constrained Devices	Ascon vs AES
14	Data-at-Rest Protection Framework	Full-disk vs file-level encryption
15	Cryptographic Algorithm Selection Framework	Rule-based vs multi-criteria selection

Public-Key Cryptography, PKI & Digital Signatures

No.	Project Topic	Suggested Two Approaches
16	RSA vs ECC for Secure Communication	RSA vs ECC
17	Secure Key Exchange System	Diffie-Hellman vs ECDH
18	Digital Signature Verification System	RSA-PSS vs ECDSA
19	Secure Digital Document Signing Platform	RSA vs ECDSA
20	Certificate Validation and Trust Management	CRL vs OCSP
21	Public-Key Authentication System	RSA vs ECC
22	Secure Certificate Management Framework	Centralized PKI vs decentralized trust

23	Cryptographic Key Rotation System	Time-based vs risk-based rotation
24	Secure Key Storage Architecture	Software vs hardware-backed storage
25	Hybrid Classical/Post-Quantum Key Exchange	ECDH vs hybrid ECDH+PQC
26	Post-Quantum Digital Signature Comparison	ML-DSA vs ECDSA
27	Post-Quantum Key Encapsulation Comparison	ML-KEM vs ECDH
28	Secure PKI for IoT Devices	PSK vs certificates
29	Digital Certificate Misuse Detection	Rule-based vs anomaly-based
30	Enterprise Trust Management	Traditional CA vs decentralized trust

Hashing, Password Security & Integrity

No.	Project Topic	Suggested Two Approaches
31	Secure Password Storage System	Argon2 vs bcrypt
32	Password Hashing Attack Resistance	PBKDF2 vs Argon2
33	File Integrity Verification	SHA-256 vs SHA-3
34	Tamper Detection for Digital Evidence	SHA-256 vs SHA-3
35	Software Integrity Verification	Hashes vs digital signatures
36	HMAC-Based API Authentication	HMAC-SHA-256 vs HMAC-SHA-3
37	Password Cracking Resistance Simulator	Dictionary vs rule-based analysis
38	Secure Password Policy Evaluation	Static vs risk-adaptive policy
39	Data Integrity Monitoring	Periodic hashing vs event-driven checks
40	Secure Configuration Integrity Checker	Hash comparison vs signed configuration

Secure Communication, TLS & Messaging

No.	Project Topic	Suggested Two Approaches
41	End-to-End Encrypted Chat Application	RSA vs ECC key establishment
42	Secure Messaging System	AES+RSA vs AES+ECC
43	Secure Email Communication Prototype	PGP-style vs S/MIME-style
44	Secure Client–Server Communication	TLS vs application-layer encryption
45	Secure File Transfer Application	SFTP/SSH vs TLS
46	Secure Group Messaging Architecture	Centralized vs distributed group keys
47	Secure Voice Communication Prototype	AES-GCM vs ChaCha20-Poly1305
48	TLS Configuration Security Analyzer	Rule-based vs vulnerability scoring
49	Secure API Communication Framework	Mutual TLS vs token authentication

50	Certificate Pinning for Secure Mobile Communication	Pinning vs conventional CA validation
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Network Security, IDS & Traffic Analysis

No.	Project Topic	Suggested Two Approaches
51	Network Intrusion Detection System	Signature vs anomaly detection
52	Machine Learning-Based IDS	Decision Tree vs Random Forest
53	Network Traffic Classification	SVM vs Random Forest
54	DDoS Attack Detection	Statistical vs ML-based
55	Port Scan Detection	Rule-based vs ML-based
56	Network Anomaly Detection	Threshold vs clustering
57	Malicious Traffic Detection	Random Forest vs XGBoost
58	Real-Time Packet Threat Detection	Signature vs behavioral
59	Network Threat Classification	Supervised vs unsupervised learning
60	Encrypted Traffic Classification	Flow-based ML vs statistical
61	DNS Anomaly Detection	Rule-based vs ML-based
62	ARP Spoofing Detection	Static ARP inspection vs anomaly detection
63	Network Reconnaissance Detection	Signature vs behavior-based
64	Botnet Traffic Detection	Flow statistics vs ML
65	Network Security Log Analysis	Rule correlation vs ML correlation

Firewall, Zero Trust & Network Defense

No.	Project Topic	Suggested Two Approaches
66	Intelligent Firewall System	Rule-based vs ML-assisted
67	Adaptive Firewall Rule Optimization	Heuristic vs genetic algorithm
68	Malicious IP Detection and Blocking	Blacklist vs ML classification
69	Automated Threat Blocking	Signature vs anomaly-based
70	Zero Trust Network Access Prototype	Traditional perimeter vs Zero Trust
71	Network Segmentation Security System	VLAN vs software-defined segmentation
72	Micro-Segmentation for Enterprise Security	Static vs dynamic policy
73	Firewall Log Analytics	Statistical vs ML-based
74	Secure Remote Access Framework	VPN vs Zero Trust access
75	Risk-Based Network Access Control	Static ACL vs adaptive access

Web, API & Application Security

No.	Project Topic	Suggested Two Approaches
76	SQL Injection Detection System	Rule-based vs ML-based
77	Cross-Site Scripting Detection	Signature vs ML-based
78	Web Application Intrusion Detection	Rules vs behavioral analysis
79	Secure API Gateway	JWT vs mutual TLS
80	API Authentication and Authorization	OAuth/OIDC vs certificates
81	Secure Web Login System	MFA vs passwordless

		authentication
82	Session Hijacking Detection	Rule-based vs behavioral
83	Malicious URL Detection	Random Forest vs neural network
84	Phishing Website Detection	Feature-based ML vs deep learning
85	Web Vulnerability Detection Tool	Signature vs ML-based

IoT, Cloud & Modern Security

No.	Project Topic	Suggested Two Approaches
86	Secure IoT Communication Framework	TLS vs DTLS
87	IoT Device Authentication	PSK vs certificates
88	IoT Intrusion Detection	Signature vs anomaly
89	Secure Smart Home Network	Centralized vs distributed security
90	Lightweight IoT Cryptography	Ascon vs AES
91	Cloud File Security Framework	Client-side vs server-side encryption
92	Cloud Access Control System	RBAC vs ABAC
93	Cloud Intrusion Detection	Signature vs ML
94	Privacy-Preserving Cloud Data Sharing	Attribute-based vs public-key encryption
95	Cloud Secrets Management	Application-managed vs centralized secrets

Advanced & Research-Oriented Security

No.	Project Topic	Suggested Two Approaches
96	Blockchain-Based Digital Identity	Blockchain vs centralized PKI
97	Blockchain-Based Certificate Verification	Blockchain vs conventional database
98	Privacy-Preserving Authentication	Zero-knowledge proofs vs conventional authentication
99	Secure Federated Learning	Differential privacy vs secure aggregation
100	AI-Based Cryptographic and Network Attack Detection	Classical ML vs deep learning

11. Final Student Checklist

- ☐ Team has 4–5 students.
- ☐ Team is divided into two sub-groups of 2–3 students.
- ☐ One common problem statement is selected.
- ☐ Two technically different approaches are defined.
- ☐ Topic and team are reported on the specified date.
- ☐ Literature review is completed.
- ☐ Both approaches are implemented/tested where applicable.
- ☐ Results and comparative metrics are recorded.

- ☐ Best approach is identified with technical justification.
- ☐ Academic report is complete.
- ☐ Draft research paper is complete for the 3-member sub-group.
- ☐ Presentation is ready.
- ☐ All members are prepared for Viva.

IMPORTANT: Any delay or lack of preparedness attributable to the students will result in deduction of marks.